

1 Graphic Process Analysis

Overview

Menu	Quality management → Process analysis → Graphic process analysis
Transaction code	gpa
Function authorization	gpa

The "graphic process analysis" is used for the process analysis in quality management.

Usage

The graphic process analysis provides a powerful graphic analysis tool for displaying process values. These are read out from the database based on time period and selection criteria. The tool also compares process values with each other. It is also possible to show the measured values along with the machine status or binary process events.

Additionally, data can be presented in a compact view. Therefore, large amounts of data can be analyzed with respect to their range and mean values.

The selected process value lines and events can also be saved as a compilation under a freely selectable name so that analyses that recur often can be called up easily.

Integration

The function displays collected measured values of the process characteristics, events and machine statuses. These values must have been previously collected in the system.

Selection criteria

The application provides the following selection criteria:

"Machine" tab

Workplace

The required workplace can be selected specifically in the "machine" field using the pool of workplaces application.

Machine status

One or several possible machine statuses can be selected. The periods of time when the selected statuses did not apply for the machine are highlighted in the graphic (gray, transparent).

"Order" tab

MES order number

Selects a specific operation (MES order number). The required MES order number can be selected specifically in the field using the pool application.

ERP batch

Selects a specific operation (MES order number)

"ERP batch" tab**Internal batch number**

Specifically selects a produced MES batch and/or produced ERP batch. The required internal batch number can be selected specifically in the field using the pool application.

Alternative batch number 1

Selects the "alternative batch number 1" at the produced batch.

General selection area**Time domain from – until**

Specifies a specific period of time for the evaluation.

Process parameters

The process parameters drop down list includes a selection of process parameters.

Tag type + tag ID

Selection of the tag type (in a user-friendly manner by a drop-down list) and the tag ID.

Automatic compression

The system specifies an automatic compression value and the compression dimension based on the selection parameters if the checkbox is checked.

Compression value

Includes the compression value. The compression value and the compression dimension summarize the individual measured values in the (compact) view.

Compression dimension

Includes the compression dimension. Seconds, minutes, hours or days can be chosen.

Consider long-term data

Long-term data may also be taken into account by clicking this checkbox.

Please note that all data used for these evaluations/reports are kept in the server memory. Consequently, it is not recommended to select too large periods and data sets.

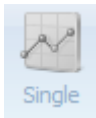
Technical background: Every time data is requested in client applications, it will be stored in the server memory and only then transferred to the client. If you require more memory space, the memory reserved for the Java application needs to be increased accordingly in the server. Please refer to the technical documentation or contact MPDV Support.

Toolbar



Show process values

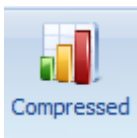
Show or hides the analog trace.



Single

The graphic shows individual process values. They are not compressed.

Please note: When switching from "compressed" to "single", the graphic will only be refreshed, once data has been requested.



Compressed

The graphic shows compressed process values.

Please note: When switching from "single" to "compressed", the graphic will only be refreshed, once data has been requested.



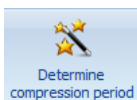
Show machine status

Shows or hides the machine status (machine time profile). It is shown in the lower section of the window.



Show events

Shows or hides the bit trace (events). It is shown in the lower section of the window.



Determine compression period

Generates a recommended compression value and the compression dimension based on the entered selection parameters and transfers the value to the selection field "compression value" and "compression dimension".

These specifications are only used if the application was set to "compressed" beforehand.

Detail applications

The graphics of the selected area are displayed in the graphic process analysis area. The buttons



can be used to insert another area for the graphic process analysis in which a new process parameter can be selected. They can also be used if an area inserted for graphic process analysis is removed.

A special feature of this graphic is the ability to zoom in and out using a keyboard and mouse. To zoom in on the graphic, press the SHIFT key and hold it down and then use the left mouse button to highlight an area of the graphic. After the key is released, this area is enlarged in the display.

To zoom out again, press and hold the ALT key down. Now the mouse pointer appears as a magnifier with a minus sign. By clicking in the graphic, the view zooms out again and the image appears smaller.

If the machine time profile is shown, the user also can carry out selections for the graphic of the machine time profile there. In the machine time profile, the machine status is always displayed.

If the events are shown, individual events can be selected there.

Display configuration

The following configuration options can be set for each chart:

Checkbox “Legend”

A legend may be shown for the data displayed in the diagram by checking this option. In case a legend is shown, axis labeling is automatically hidden to save space.

Checkbox “Show scale”

The scale of the diagram can be shown or hidden by selecting this checkbox.

Checkbox “Show characteristic name”

This checkbox specifies that the long characteristic names are shown for the legend and axis labeling instead of the short ID. Please note: labeling is not updated until data has been refreshed!